O17e - Diffraction

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Abstract

This experiment investigates the diffraction and interference of monochromatic laser light through single and double slits in both the Fraunhofer and Fresnel regimes. Intensity profiles were recorded using a CCD camera and analyzed to extract slit parameters via spatial and frequency-domain fitting techniques. In the Fraunhofer regime, the single-slit diffraction pattern yielded a slit width of $b \approx 206.7\,\mu$m. For the double slit, three independent methods consistently determined the slit width and separation as $b \approx 239\,\mu$m and $g \approx 364\,\mu$m, with correlation coefficients of $r^2 > 0.99$ in all fits. In the Fresnel regime, the evolution of the fringe structure with decreasing slit-to-screen distance was accurately reproduced by numerical Fresnel modeling using the same fitted values of b and g, achieving best-fit r^2 values of up to 0.96. While the measured parameters slightly overestimated the nominal values ($b_{\text{exp}} = 200\,\mu$m, $g_{\text{exp}} = 300\,\mu$m) for the double-slit analysis, the internal consistency between methods confirms the reliability of the results. The experiment demonstrates the predictive power of wave optics and its ability to bridge the near- and far-field regimes in diffraction phenomena.

1 Introduction

The phenomenon of diffraction and interference lies at the heart of wave optics and provides direct evidence of the wave nature of light. When coherent light encounters apertures or obstacles whose dimensions are comparable to its wavelength, it bends around the edges and produces characteristic intensity distributions that depend sensitively on the geometry of the system and the observation conditions. These patterns encode information about the structure of the aperture and the wavelength of the light, and their systematic study forms a fundamental part of experimental optics.

In particular, the double-slit experiment serves as a classical and conceptually powerful demonstration of wave interference. When monochromatic light passes through two closely spaced slits, each slit acts as a secondary source of spherical wave-fronts. The superposition of these waves leads to an interference pattern on a distant screen, characterized by alternating bright and dark fringes. The detailed form of this pattern is shaped not only by the spacing between the slits but also by the finite width of each slit, which introduces additional diffraction effects.

The observed pattern depends strongly on the distance between the slit and the observation screen. In the limit where this distance is large compared to the square of the slit width divided by the wavelength, the pattern is well described by Fraunhofer diffraction theory, which assumes planar wave-fronts and results in analytically tractable expressions involving sinc and cosine functions. In contrast, when the observation is made at distances comparable to the slit dimensions, the wavefront curvature becomes significant, and the Fresnel regime must be considered. This near-field diffraction is more complex and requires numerical evaluation of Fresnel integrals to predict the intensity distribution.

By studying both the Fraunhofer and Fresnel regimes in controlled experimental conditions, one gains a deeper understanding of the transition between near- and far-field optics and the mathematical frameworks that describe them. Moreover, the precise measurement and modeling of diffraction patterns provide a powerful method for determining geometric parameters such as slit width and separation, and for validating the theoretical foundations of wave optics.

2 Theoretical Analysis

2.1 Fourier Transformation and Fraunhofer Pattern of a Single and Double Slit
(Task 0)

2.1.1 Theoretical Derivation

Single Slit:
The transmission function of a single slit of width b centered at the origin is given by:

$$f(x') = \begin{cases} 1, & |x'| < \dfrac{b}{2} \\ 0, & \text{otherwise} \end{cases}$$

The Fraunhofer diffraction pattern in the far field corresponds to the Fourier transform of $f(x')$. Using the physics convention for the Fourier transform:

$$F(k_x) = \int_{-\infty}^{\infty} f(x') \cdot e^{-ik_x x'}\, dx'$$

we obtain:

$$F(k_x) = \int_{-b/2}^{b/2} e^{-ik_x x'}\, dx' = b \cdot \operatorname{sinc}\left(\frac{k_x b}{2}\right)$$

where $\operatorname{sinc}(u) = \frac{\sin u}{u}$. The intensity recorded by the camera is proportional to the squared modulus of the amplitude:

$$I(k_x) \propto |F(k_x)|^2 = b^2 \cdot \operatorname{sinc}^2\left(\frac{k_x b}{2}\right)$$

For small diffraction angles α, the lateral momentum component is $k_x = k\sin(\alpha)$. This matches the theoretical expression given in Eq. (7) of [1].

Double Slit:
For two slits of equal width b separated by a center-to-center distance g, the transmission function becomes:

$$f(x') = \begin{cases} 1, & -\frac{g+b}{2} < x' < -\frac{g-b}{2} \\ 1, & \frac{g-b}{2} < x' < \frac{g+b}{2} \\ 0, & \text{otherwise} \end{cases}$$

Rewrite it as a sum of two shifted single-slit rectangles:

$$f(x') = \text{rect}\left(\frac{x' + g/2}{b}\right) + \text{rect}\left(\frac{x' - g/2}{b}\right)$$

Using the shift (displacement) theorem, the Fourier transform is

$$F(k_x) = \text{sinc}\left(\frac{k_x b}{2}\right)\left[e^{+ik_x g/2} + e^{-ik_x g/2}\right] \tag{1}$$

$$= 2b\,\text{sinc}\left(\frac{k_x b}{2}\right)\cos\left(\frac{k_x g}{2}\right). \tag{2}$$

Thus, the intensity becomes:

$$I(k_x) \propto |F(k_x)|^2 = 4b^2 \cdot \text{sinc}^2\left(\frac{k_x b}{2}\right) \cdot \cos^2\left(\frac{k_x g}{2}\right)$$

which is Eq. (8) of [1]. The diffraction envelope is determined by the single-slit sinc-squared term, while the cosine-squared term produces the interference fringes.

2.1.2 Numerical Confirmation via FFT and the Convolution Theorem

To confirm the convolution theorem numerically, we first calculate the discrete Fourier transform of the sinc function and its square.
We define the function:

$$F(k) = \text{sinc}(2\pi k) = \frac{\sin(2\pi k)}{2\pi k}$$

The FFT of $F(k)$ yields a rectangular function in real space, as expected from the analytical Fourier transform of a sinc function, we observe a rectangular profile in real space. The numerical FFT yields a rectangular pulse of width approximately 1.
Next, we square the function element-wise:

$$F^2(k) = \text{sinc}^2(2\pi k)$$

Here, the FFT yields a triangular function. This is consistent with the convolution theorem, which states:

$$\mathcal{F}\{f \cdot f\} = F * F$$

That is, squaring the sinc function in k-space results in convolving the corresponding rectangular functions in real space — the outcome being a triangular shape.
For clarity, the x-domain was truncated to $[-30, 30]$, which retains the central structure of the patterns while improving readability. This omits the very far-field tails, which have minimal contribution to the overall intensity.

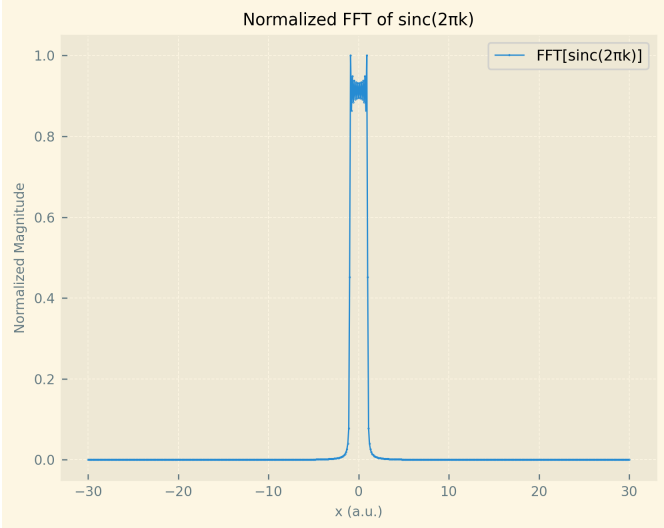

Figure 1: (Fast-Fourier-Transform of $F(k) = \text{sinc}(2\pi k)$

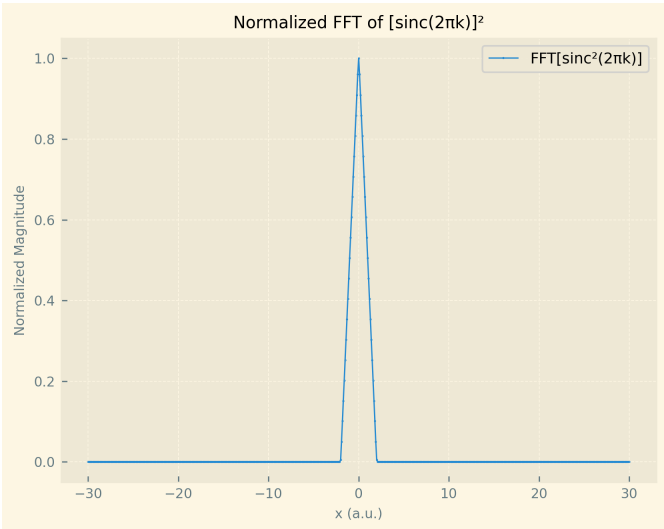

Figure 2: Fast-Fourier-Transform of $F(k)^2 = \text{sinc}^2(2\pi k)$

2.2 Single Slit Diffraction Pattern

When a monochromatic wavefront encounters a narrow slit, it diffracts and spreads out. The pattern observed on a screen depends on the distance between the slit and the observation point. Two main regimes are distinguished:

2.2.1 Fraunhofer Diffraction (Far-Field)

In the far-field limit — known as the Fraunhofer regime — the diffracted wavefronts can be approximated as planar. The condition for Fraunhofer diffraction is that the distance from the slit to the observation plane (or focal plane of a lens) is large compared to the square of the slit width divided by the wavelength:

$$z \gg \frac{b^2}{\lambda}.$$

The resulting intensity pattern for a single slit of width b is given by:

$$I(x) = I_0\left(\frac{\sin(\beta)}{\beta}\right)^2, \quad \text{with} \quad \beta = \frac{\pi b x}{\lambda f},$$

Where:

x : lateral position on the screen relative to the center
λ : wavelength of the light,
f : focal length of the imaging lens,
I_0 : central maximum intensity.

The central bright fringe is the widest, and intensity minima occur where $\beta = n\pi$, i.e., at positions where:

$$x_n = \frac{n\lambda f}{b}, \quad n \in \mathbb{Z} \setminus \{0\}. \tag{3}$$

This pattern is characterized by a central peak and diminishing side lobes, and its envelope is governed by the sinc-squared function.
For a single slit of width b, the aperture function is a rectangular function:

$$T(x') = \begin{cases} 1, & |x'| \leq \frac{b}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Its Fourier transform yields:

$$F(k_x) = b \cdot \operatorname{sinc}\left(\frac{k_x b}{2}\right), \quad k_x = \frac{2\pi x}{\lambda f}. \tag{4}$$

The observed intensity is proportional to the squared modulus:

$$I(k_x) \propto |F(k_x)|^2 = b^2 \cdot \operatorname{sinc}^2\left(\frac{k_x b}{2}\right).$$

This interpretation shows that the far-field diffraction pattern contains spatial frequency information about the aperture.

2.2.2 Fresnel Diffraction (Near-Field)

In the Fresnel regime — the near-field limit — the distance between slit and screen is finite, and the curvature of the wavefront must be accounted for. The field at the observation point is computed using the Fresnel approximation to the Kirchhoff diffraction integral:

$$\Psi(x; z) \propto \int_{-b/2}^{b/2} \exp\left(i\frac{\pi}{\lambda z}(x - x')^2\right) dx'.$$

This leads to the intensity:

$$I(x; z) \propto |C(u_+) - C(u_-) + i[S(u_+) - S(u_-)]|^2, \tag{5}$$

Where:

- $u_\pm = \sqrt{\frac{2}{\lambda z}}\left(x \pm \frac{b}{2}\right)$,

- $C(u)$ and $S(u)$ are the Fresnel cosine and sine integrals,

$$C(u) = \int_0^u \cos\left(\frac{\pi}{2}t^2\right) dt, \quad S(u) = \int_0^u \sin\left(\frac{\pi}{2}t^2\right) dt.$$

The Fresnel pattern evolves with distance, gradually transitioning into the Fraunhofer pattern at larger z.
The transition from Fresnel to Fraunhofer diffraction is continuous. A commonly used parameter to characterize this transition is the Fresnel number:

$$F = \frac{b^2}{\lambda z}. \tag{6}$$

For $F \gg 1$: we are deep in the Fresnel regime.
For $F \ll 1$: the Fraunhofer approximation applies.

This number provides a quantitative basis for choosing the correct theoretical model in diffraction problems.

2.3 Double-Slit Diffraction

2.3.1 Fraunhofer Diffraction (Far-Field)

In the far-field or Fraunhofer regime, light diffraction through a double slit can be described by the interference of two coherent wavefronts, each modulated by the diffraction envelope of a finite-width slit. The resulting intensity pattern at a distant observation screen is governed by the principle of superposition and the scalar diffraction theory under the paraxial approximation.

The relation between the position of the fringe and the fringe order is give by:

$$x_n = \frac{n\lambda f}{b}, \quad n \in \mathbb{Z} \setminus \{0\}. \tag{7}$$

The intensity distribution $I(x)$ as a function of lateral position x in the observation plane is given by:

$$I(x) = I_0 \left(\frac{\sin(\beta)}{\beta}\right)^2 \cos^2(\delta), \tag{8}$$

where

$$\beta = \frac{\pi b x}{\lambda f}, \quad \delta = \frac{\pi g x}{\lambda f}.$$

Here, b is the slit width, g is the center-to-center separation between the two slits, λ is the wavelength of the incident light, and f is the focal length of the imaging lens. The sinc^2 function defines the diffraction envelope arising from the finite width of each slit, modulating the overall shape of the intensity profile. The $\cos^2$ term describes the interference fringes caused by the path difference between light from the two slits.. The intensity maxima are periodic in x, with fringe spacing inversely related to the slit separation, and the envelope determines the overall width of the diffraction pattern.

This expression assumes a plane wave incident on the aperture and an observation screen effectively at infinity, or equivalently, a Fourier plane formed by a lens at distance f from both the aperture and the detector. It is a standard result of the Fourier optics treatment of apertures and is applicable in situations where the slit dimensions are small compared to the propagation distance or the focal length.

In the spatial frequency domain, the intensity distribution can also be expressed in terms of the transverse wavevector k_x, defined by $k_x = \frac{2\pi x}{\lambda f}$. The intensity profile in k_x-space becomes:

$$I(k_x) = I_0 \cdot \operatorname{sinc}^2\left(\frac{b k_x}{2}\right) \cdot \cos^2\left(\frac{g k_x}{2}\right) + C_0, \tag{9}$$

where the sinc-squared function represents the diffraction envelope and the cosine-squared term represents the interference fringe structure. The constant C_0 accounts for a uniform background intensity. This formulation arises from the Fourier transform of the double-slit aperture

function and highlights the fundamental duality between real-space and frequency-space diffraction patterns.

2.3.2 Fresnel Diffraction (Near-Field)

When the propagation distance between the aperture and the observation screen is comparable to the square of the slit width divided by the wavelength, the Fraunhofer approximation no longer holds. In this near-field or Fresnel regime, the wavefront curvature must be accounted for, and the diffraction pattern becomes sensitive to the propagation distance. The appropriate theoretical treatment is based on the Fresnel-Kirchhoff diffraction integral.

Under the scalar and paraxial approximations, the electric field at the observation plane due to a double-slit aperture can be described using Fresnel integrals. The complex amplitude from a single rectangular slit is given by the difference of Fresnel integrals evaluated at the edges of the slit. For a double slit, the total field is a coherent sum of the contributions from both slits, and the intensity $I(x;z)$ at the screen is given by:

$$I(x;z) \propto \left| F\left(\frac{g+b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{g+b-2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b+2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b-2x}{\sqrt{2\lambda z}}\right) \right|^2 \tag{10}$$

where $F(u) = C(u) + iS(u)$ is the complex Fresnel integral composed of the cosine integral $C(u)$ and sine integral $S(u)$. The argument of the integrals scales with the inverse square root of the propagation distance z, such that the spatial extent and structure of the diffraction pattern evolve as the distance changes.

This formulation captures both the self-diffraction of individual slits and the interference between them, while accounting for phase differences arising from the curvature of the spherical wavefront. The diffraction pattern exhibits a gradual transition from the complex oscillatory features of the near field to the more symmetric and regularly spaced fringes of the far-field Fraunhofer pattern.

The transition between Fresnel and Fraunhofer regimes is governed by the dimensionless Fresnel number (Eq. (6)), which quantifies the importance of near-field effects. Large Fresnel numbers ($F \gg 1$) correspond to the deep Fresnel regime with significant curvature and interference effects, while small Fresnel numbers ($F \ll 1$) indicate that the pattern approaches the far-field limit. The Fresnel formalism is essential for accurately modeling diffraction at finite distances and plays a central role in optical systems lacking imaging lenses.

3 Experimental Analysis

3.1 Single-Slit Diffraction
(Task I)

Fraunhofer Regime

The experimental arrangement in Fig. 3 was designed to observe the Fraunhofer diffraction pattern of a single slit. A coherent red laser beam with a wavelength of

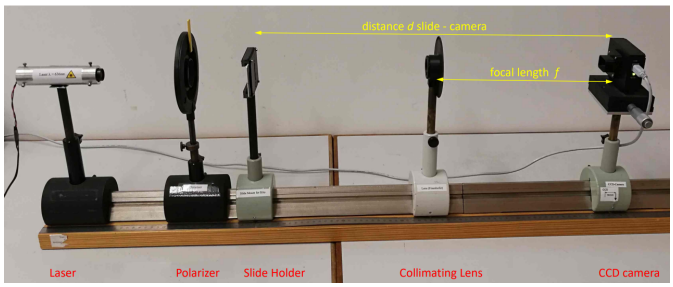

Figure 3: Experimental Setup to Observe Fraunhofer Diffraction Pattern

Source: [2]

$\lambda = 636\,\mathbf{nm}$ was directed through a narrow single slit of width $b_{exp} = 200\,\mu$m. The diffracted light was then passed through a converging lens with a focal length $f = 301.7\,$mm, which served to project the far-field diffraction pattern onto a CCD camera positioned in the focal plane.

The CCD camera captured the resulting intensity distribution along a linear array of pixels, each with a physical width of $14\,\mu$m . The data was saved as intensity versus pixel number. For the purpose of analysis, this pixel scale was converted to a lateral position in millimeters, centered about the maximum intensity.

This setup allowed the application of both direct spatial analysis (Task 1a) and Fourier-domain methods (Task 1b) to extract information about the slit width and verify the theoretical predictions.

Fresnel Regime

to explore near-field (Fresnel) diffraction effects, the setup showing Fig. 3 was modified. The same monochromatic red laser and single slit were used, but the converging lens was removed. The CCD camera was placed at a fixed position on an optical rail, while the slit was moved incrementally along the rail to vary the distance z between the slit and the camera.

The camera remained fixed at the 137.6 cm mark. The slit was positioned at three different locations (115 cm, 125 cm, and 130 cm), creating effective propagation distances of 22.6 cm, 12.6 cm and 7.6 cm. For each slit position, the camera recorded the corresponding near-field diffraction pattern.

These data sets were later compared to theoretical Fresnel predictions, using the known slit width $b_{exp} = 0.2\,$mm, to examine how the diffraction pattern evolved with distance and how well it matched the Fresnel integral-based model.

3.1.1 Measurement and Analysis

Fraunhofer Regime

The Fraunhofer diffraction pattern was first analyzed in real space by identifying the positions of eight consecutive intensity minima symmetrically located around the central maximum. The difference in their positions provided the values x_n corresponding to each diffraction order n.

A linear regression of x_n against n was performed using the relation provided by Eq. (3):

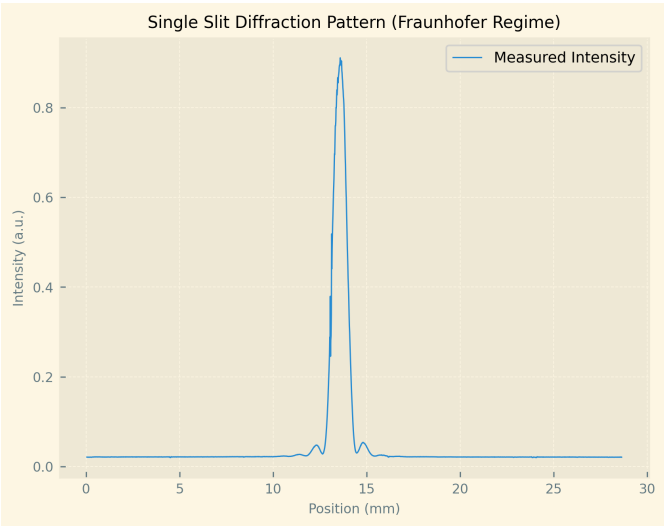

Figure 4: Intensity Profile of Single-Slit Diffraction in the Fraunhofer Regime

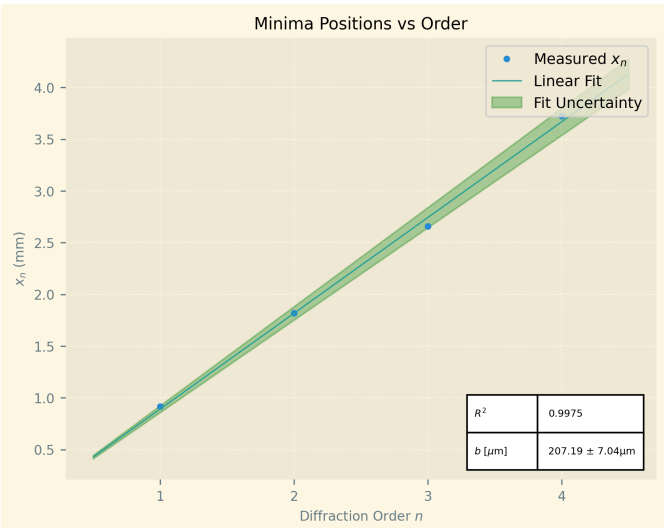

Figure 5: Fitted Minima Position vs. Order (x_n vs. n) Plot

From the slope of the fit, the slit width was extracted as $b_1^{SS} = (207.19 \pm 7.04)\,\mu m$ (where SS refers to Single-Slit). This value compares well with the experimental value of the slit width $b_{exp} = 200\,\mu m$. The high coefficient of determination of $r^2 = 0.9975$ also indicates excellent agreement with the theoretical model.

Next, the measured intensity distribution was transformed into spatial frequency domain via a Fourier transform. The real-space axis was first converted to wave-vector space using the k_x relation in Eq. (4).

The magnitude of the Fourier-transformed intensity was then fitted to the model:

$$I(k_x) \propto \mathrm{sinc}^2\left(\frac{bk_x}{2}\right),$$

allowing a second independent estimate of the slit width b.

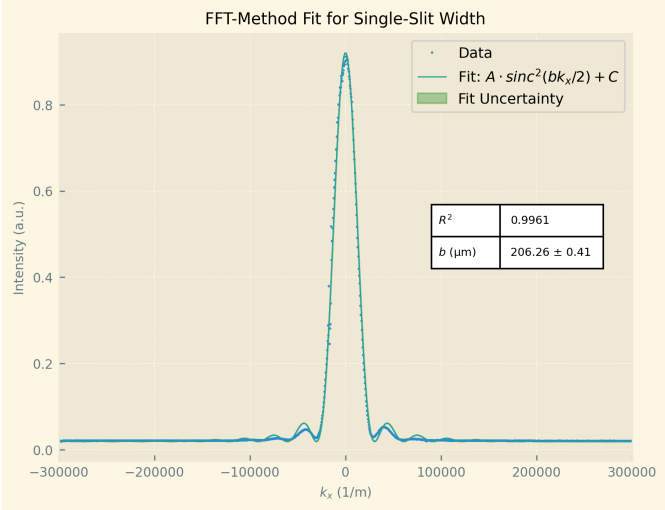

Figure 6: Fitted Fast Fourier Transform

The fit provided slit width $b_2^{SS} = (206.26 \pm 0.41)\,\mu m$ and a coefficient of correlation of $r^2 = 0.9961$, confirming the consistency of the Fourier-domain model with the observed data. The obtained slit width value lies within the the uncertainty of the value obtained from the linear regression in Fig. 5 and is close to the experimental value of b.

The slit widths extracted from each method showed excellent agreement, validating both the spatial and spectral analysis techniques and supporting the theoretical prediction of the Fraunhofer diffraction envelope.

Fresnel Regime

For the Fresnel diffraction measurements, intensity profiles were recorded at three different slit-to-camera distances 22.6 cm, 12.6 cm and 7.6 cm. Each intensity profile was fitted to the theoretical Fresnel diffraction formula (Eq. (5)) based on the known slit width $b = 0.2\,\mathrm{mm}$.

For each profile, the fit parameters included an amplitude scaling factor A and a background offset C_0 which accounted for ambient/detector background light. These were obtained in arbitrary units (a.u.). The match between theory and measurement was quantified using r^2 values:

- Slit at 7.6 cm: The fit yielded $r^2 = 0.9834$, with amplitude $A = 0.584 \pm 0.002$ and background $C_0 = 0.019$.
- Slit at 12.6 cm: The fit yielded $r^2 = 0.9943$, with $A = 0.389 \pm 0.001$, $C_0 = 0.024$.
- Slit at 22.6 cm: The fit yielded $r^2 = 0.9824$, with $A = 0.179 \pm 0.001$, $C_0 = 0.024$.

In all cases, the fitted Fresnel curves closely tracked the measured intensity oscillations. The measured peaks aligned well with those predicted by the Fresnel model, indicating accurate phase and geometric calibration. The

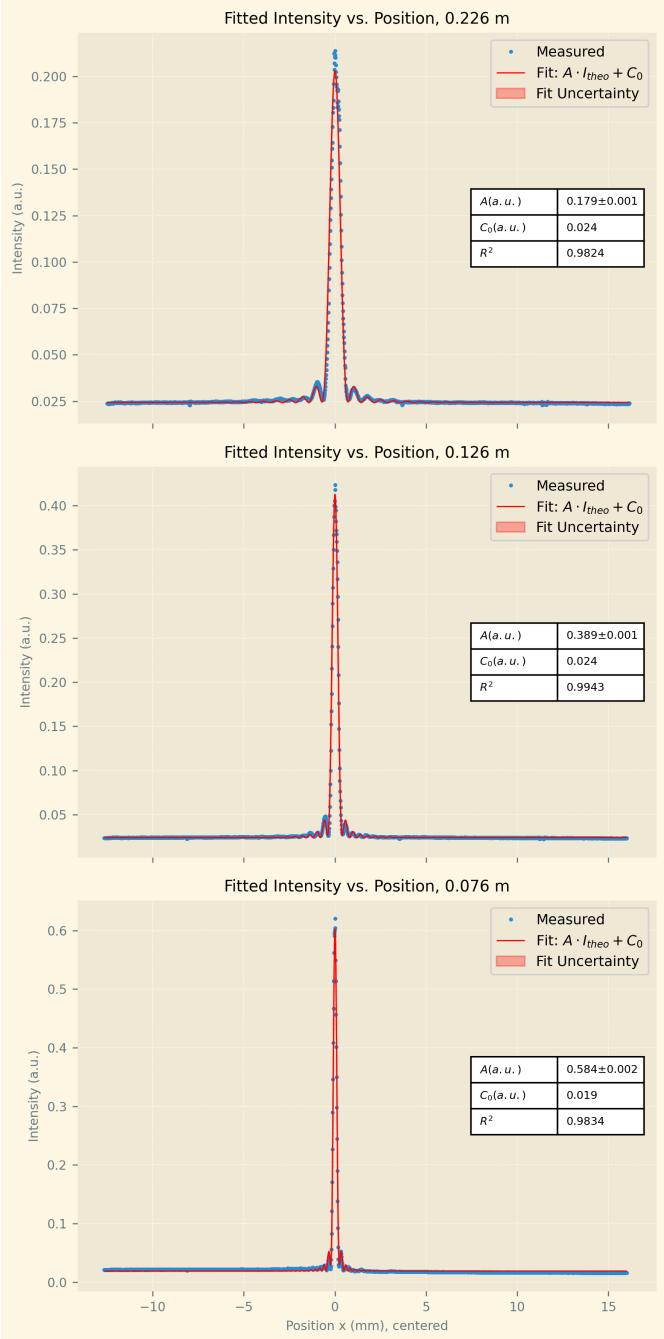

Figure 7: Fitted Intensity Profiles at Three Different Slit-Camera Distances (22.6 cm, 12.6 cm and 7.6 cm)

visibility of the fringes was preserved throughout the range, with good contrast particularly in the central regions. Slight deviations near the edges were likely due to lower signal-to-noise ratios.The fit quality improved with increasing distance, as expected, because the patterns transition from near-field to far-field regimes. The overall envelope of the experimental data matched the theoretical predictions, especially at longer distances where the Fresnel pattern begins to approximate the Fraunhofer limit. For $z = 22.6\,\text{cm}$, the profile began to resemble the Fraunhofer envelope, consistent with the theory. Quantitatively, the Fresnel number $F = b^2/(\lambda z)$ drops from ≈ 0.83 at 7.6 cm to ≈ 0.28 at 22.6 cm. Since $F \ll 1$ in the latter case, the diffraction pattern converges to the Fraunhofer limit—explaining the observed resemblance.

3.2 Double-Slit Diffraction
(Task II)

The investigation of the diffraction behavior of a double slit was carried out under two different propagation conditions: the far-field or Fraunhofer regime, and the near-field or Fresnel regime. In both cases, the experimental apparatus was adapted from the single-slit configuration, with modifications made specifically to introduce and characterize the effects of the slit separation.

The pixel resolution and magnification factors remained unchanged between the two regimes, enabling direct comparison of the intensity distributions across configurations. For all measurements, care was taken to center the optical pattern on the camera and subtract any background offset in the recorded data. The double-slit parameters—slit width and separation—were kept constant throughout the experiment, ensuring that any variation in the observed patterns could be attributed purely to the change in diffraction regime and propagation distance.

Fraunhofer Regime

The same setup was used as shown in Fig. 3 except a double-slit aperture was mounted at the position where the single-slit aperture was previously used. The two slits in the aperture were of equal width of $b_{exp} = 200\,\mu\text{m}$ and separated by a finite center-to-center distance of $g_{exp} = 300\,\mu\text{m}$. In the Fraunhofer regime, a converging lens of focal length $f = 301.7\,\text{mm}$ was used to image the diffraction pattern into the far-field. The CCD camera was placed in the focal plane of the lens. The laser source ($\lambda = 636\,\text{nm}$), lens, and camera configuration remained identical to the previous arrangement used for single-slit diffraction. The alignment was carefully maintained to ensure that the optical axis passed symmetrically through both slits, and the resulting interference pattern was recorded by the camera with high spatial resolution.

Fresnel Regime

In the near-field or Fresnel regime, the imaging lens was removed entirely. Instead, the diffraction pattern was allowed to propagate freely from the double-slit to the CCD sensor. The camera was fixed at a constant position near the end of the optical rail, while the double-slit aperture was translated to several positions (with slit-camera distances: 42.6 cm, 32.6 cm and 22.6 cm) along the rail to vary the propagation distance. This allowed the observation of how the interference and diffraction structure evolved as the observation screen approached the aperture. The choice of slit positions was made to span a range of distances where Fresnel effects were expected to be significant. In each configuration, the intensity profile was recorded over a sufficiently wide spatial domain to capture both the central and lateral fringe structures.

3.2.1 Measurement and Analysis

Fraunhofer Regime

To characterize the double-slit interference pattern in the Fraunhofer regime, the spatial positions of selected interference minima were identified from the experimental

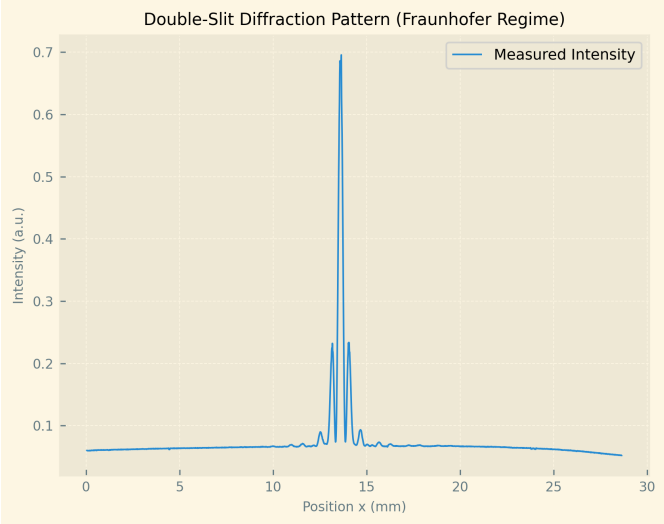

Figure 8: Intensity Profile of Double-Slit Diffraction in the Fraunhofer Regime

intensity distribution. The measurement was challenged by a relatively high noise level in the recorded signal, which rendered many of the weaker minima difficult to localize reliably. To address this, the analysis focused only on the most prominent and clearly discernible minima. This reduced the total number of usable minima, but allowed for a cleaner extraction of the spatial fringe periodicity.

The distances between symmetrically located minima around the central maximum were calculated as $2x_n$, and the corresponding fringe order n was assigned incrementally. These values were plotted to evaluate the linear relationship between fringe spacing and diffraction order shown in Fig. 9, as expected from the theoretical model give by Eq. (7)

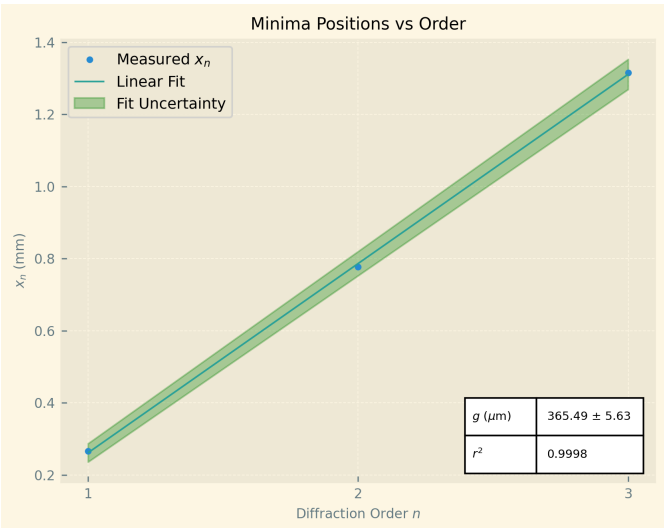

Figure 9: Fitted Plot of Minima Position against Order (x_n vs. n)

Linear regression of the data yielded $g_1^{DS} = (365.49 \pm 5.63)\,\mu$m (where DS refers to Double-Slit) and $r^2 = 0.9998$. The value of the slit separation is notably higher than the nominal experimental slit separation $g_{exp} = 300\,\mu$m, indicating a deviation of approximately 22%. The discrepancy likely originates from uncertainty in locating the interference minima due to overlapping side-lobe structures and signal noise, compounded by the relatively small number of data points used for regression. Despite this, the very high coefficient of determination supports the internal consistency of the measured values. The results suggest that the spatial periodicity of the interference fringes aligns well with theoretical expectations, even if the exact value of g is slightly overestimated.

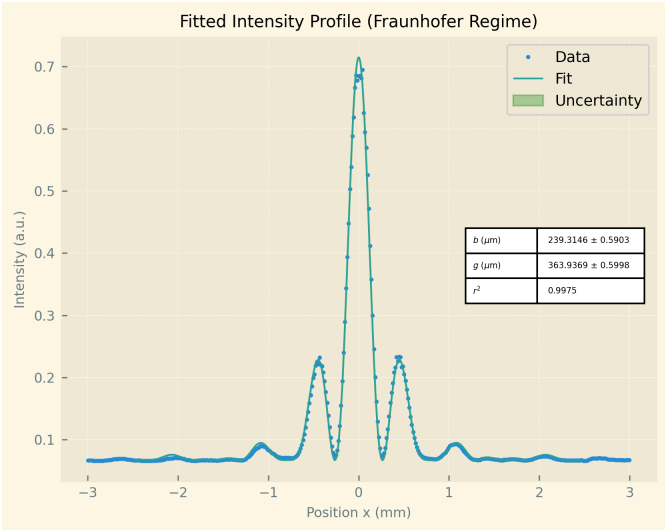

Figure 10: Fitted Fraunhofer Diffraction Pattern Data

Using another approach, the entire spatial intensity distribution of the double-slit Fraunhofer pattern was fitted directly using the analytical model for two slits of width b and separation g provided by Eq. (8). The functional form for the intensity profile incorporated both the single-slit diffraction envelope and the double-slit interference modulation, with the model parameters adjusted to best match the observed data in the central region of the pattern.

The resulting plot is shown in Fig. 10. The range of the fitting was limited to ± 3 mm to obtain a more precise fitting. The nonlinear regression fit yielded $b_2^{DS} = (239.31 \pm 0.59)\,\mu$ m and $g_2^{DS} = (363.94 \pm 0.60)\,\mu$m with $r^2 = 0.9975$. Both fitted parameters exhibit low uncertainties and a the high r^2 value reflects the quality of the fit and the suitability of the theoretical model for describing the observed data.

Comparing these results to the nominal values provided by the slit manufacturer, there is a moderate overestimation. The experimental slit separation and width were specified as $g_{\text{exp}} = 300\,\mu$m and $b_{\text{exp}} = 200\,\mu$m respectively, indicating deviations of approximately 21% for g and 20 % for b. This systematic difference may be attributed to manufacturing tolerances, misalignment, or possible calibration errors in the setup. It is also notable that the value of g obtained here is in close agreement with that determined in the previous method based on minima positions ($g_1^{DS} = 365.49 \pm 5.63\,\mu$m), reinforcing the internal consistency of the experimental results.

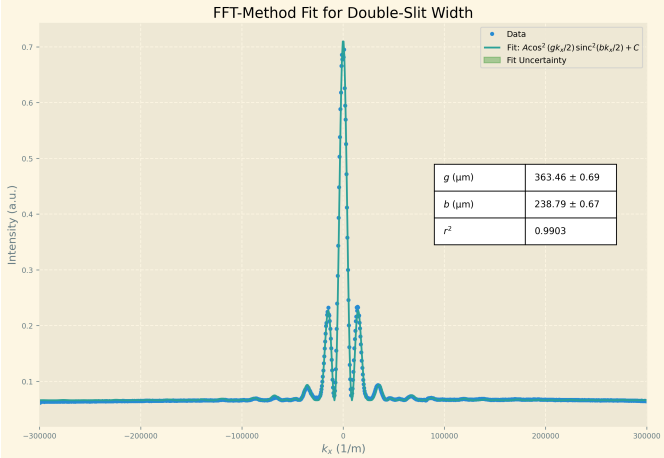

Figure 11: Fitted Fast Fourier Transform

In this final Fraunhofer analysis, the experimental intensity distribution was transformed from real space to the spatial frequency domain by expressing the intensity as a function of k_x, the wave-vector component conjugate to position. This was achieved by centering the data, converting pixel positions to wave-vector using the system's geometry, and then directly fitting the analytical double-slit intensity profile in the k_x domain. The fitted model takes the form described by Eq. (9).

According to Fig. 11 fitting the model to the experimental data yielded parameter values of $b_3^{DS} = 238.79 \pm 0.67\,\mu$m and $g_3^{DS} = 363.46 \pm 0.69\,\mu$m, with an associated coefficient of determination $r^2 = 0.9903$. These results are fully consistent with those obtained in the previous fitting approaches: the slit width and separation fall within the uncertainty range of the direct real-space fit in Fig. 10 as well as the value obtained from the minima analysis in Fig. 9. All fitted values are systematically higher than the manufacturer's nominal parameters ($b_{\text{exp}} = 200\,\mu$m, $g_{\text{exp}} = 300\,\mu$m), but they show excellent agreement with one another.

The high r^2 value underscores the quality of the fit and the suitability of the Fourier-space model for capturing the main features of the observed pattern. The consistency of the results across all three analytical methods reinforces the reliability of the measurement and the internal coherence of the experimental procedure. The systematic deviation from nominal values is likely attributable to cumulative effects of slit manufacturing tolerances, possible misalignment, or subtle calibration offsets in the optical setup, as previously discussed.

Fresnel Regime

To investigate the near-field diffraction behavior of a double slit, intensity profiles were recorded at three distinct slit-to-camera distances: $z = 42.6$ cm, $z = 32.6$ cm, and $z = 2.26$ cm. These distances correspond to increasingly pronounced Fresnel effects, supported by Eq. (6) , transitioning from the onset of near-field interference into regimes where the wavefront curvature plays a dominant role in shaping the observed pattern.

The intensity distributions recorded at each distance were centered and normalized before being fitted to the full theoretical Fresnel diffraction expression for a double slit. The model used no free parameters for the slit geometry: the values for the slit width and slit separation used were averages of the parameter values obtained from the Fraunhofer analysis. Only a global amplitude and a constant background offset were included as fit parameters. The Fresnel integrals were evaluated numerically using the standard definition provided by Eq. 10.

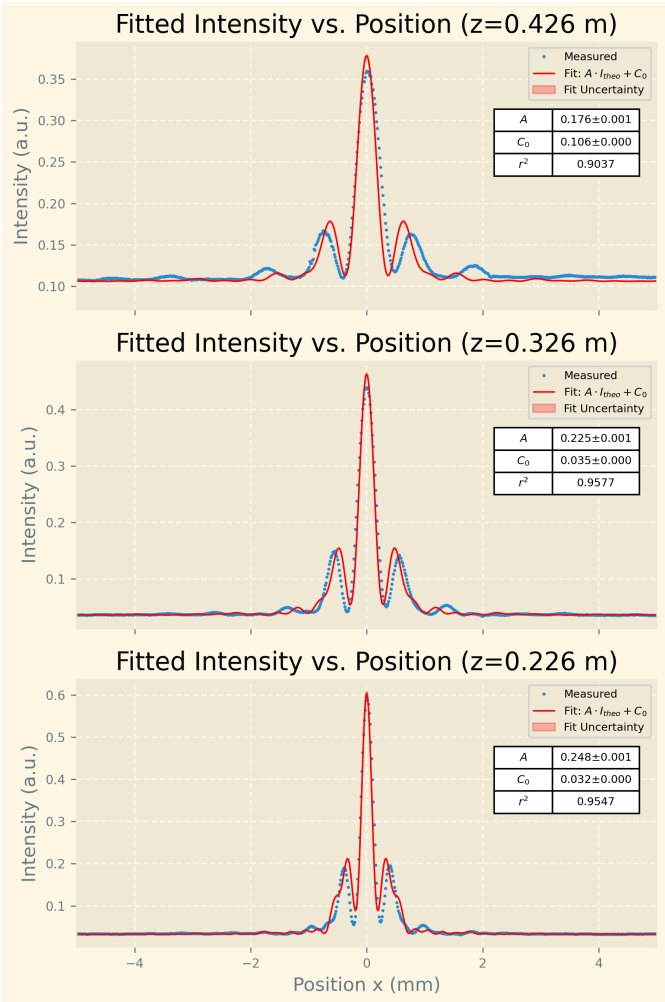

Figure 12: Fitted Intensity Profiles at Three Different Slit-Camera Distances (42.6 cm, 32.6 cm and 22.6 cm)

The measured intensity profiles for all three distances exhibited the expected qualitative features: a central fringe flanked by symmetric side lobes with increasing fringe density and contrast as the slit was moved further from the camera. As z decreased from 42.6 cm to 22.6 cm: the fringes became sharper, more localized, and the oscillatory structure was denser; the central peak narrowed, and the side-lobe modulation became more complex. This indicated that near-field contributions (Fresnel terms) are dominating more of the pattern formation.The agreement between the theoretical prediction and the measured data was consistently good across all three datasets. The coefficient of determination was highest for the intermediate distance at $z = 32.6$ cm with $r^2 = 0.9577$, followed closely by the shortest distance at $z = 22.6$ cm

with $r^2 = 0.9547$. Even the longest propagation distance ($z = 42.6\,\text{cm}$)—which borders on the Fraunhofer regime—still showed a reasonable match with the theoretical prediction, though the fit quality dropped slightly to $r^2 = 0.9037$. This decrease is likely due to the interference fringes becoming more compressed and less resolved at greater distances, which affects the sensitivity of the fit to small deviations in fringe position.

4 Discussion

4.1 Single-Slit

The diffraction measurements performed in Task 1 demonstrate strong quantitative agreement between theoretical predictions and experimental data across both Fraunhofer and Fresnel regimes.

In the Fraunhofer regime, two independent approaches were employed to extract the slit width: one using real-space minima positions (Task 1a), and the other via Fourier transformation (Task 1b). The slit width obtained from the real-space analysis was$b_1^{SS} = (207.19 \pm 7.04)\,\mu\text{m}$, while the Fourier-domain fit yielded $b_2^{SS} = (206.26 \pm 0.41)\,\mu\text{m}$. Both values lie within a narrow margin of the nominal slit width $b_{\text{exp}} = 200\,\mu\text{m}$, and their mutual consistency validates the experimental approach. Moreover, the high coefficients of determination — $r^2 = 0.9975$ and $r^2 = 0.9961$, respectively — confirm that the theoretical sinc^2 profile accurately models the observed intensity distribution. The minor systematic deviation could stem from practical uncertainties such as mechanical positioning of the slit or pixel calibration, but remains well within acceptable bounds.

In the Fresnel regime, the recorded intensity distributions at distances $z = 7.6\,\text{cm}$, $12.6\,\text{cm}$, and $22.6\,\text{cm}$ showed excellent qualitative and quantitative agreement with the Fresnel integral model. The r^2 values for the fits were all above 0.98, with the best agreement at $z = 12.6\,\text{cm}$ ($r^2 = 0.9943$), reflecting the high sensitivity of near-field patterns to both noise and alignment. Notably, peak positions were well-matched to theoretical predictions across all distances, fringe visibility was highest in the mid-range distance (12.6 cm), consistent with optimal interference contrast in the Fresnel region and the envelope shape increasingly resembled the Fraunhofer sinc^2 profile as the distance increased, confirming the expected smooth transition from near- to far-field behavior.

Overall, the consistency of the extracted parameters with known values, the fidelity of the fits, and the matching of qualitative features (fringes, envelopes, and asymptotic behavior) demonstrate that both scalar diffraction models (Fraunhofer and Fresnel) accurately describe the behavior of light passing through a single slit under the specified experimental conditions.

4.2 Double-Slit

The diffraction and interference effects observed in the double-slit setup provided a clear demonstration of the distinct behaviors in the Fraunhofer and Fresnel regimes. The far-field measurements showed high consistency between the theoretical models and experimental data, particularly in terms of fringe periodicity, overall envelope shape, and the extracted geometric parameters.

In the Fraunhofer regime, three different analysis methods yielded consistent results for the slit width and separation. The linear fit to the positions of intensity minima resulted in a slit separation of $g_1^{DS} = (365.49 \pm 5.63)\,\mu\text{m}$, substantially larger than the nominal $g_{\text{exp}} = 300\,\mu\text{m}$. A direct nonlinear fit to the full spatial intensity distribution gave $g_2^{DS} = (363.94 \pm 0.60)\,\mu\text{m}$ and $b_2^{DS} = (239.31 \pm 0.59)\,\mu\text{m}$, while the frequency-domain fit yielded nearly identical values of $g_3^{DS} = (363.46 \pm 0.69)\,\mu\text{m}$ and $b_3^{DS} = (238.79 \pm 0.67)\,\mu\text{m}$. All three fits showed very high correlation coefficients, with $r^2 \geq 0.9903$, indicating an excellent agreement with the theoretical Fraunhofer expression. The discrepancy between the fitted and nominal values—roughly 20% overestimation—likely stems from cumulative effects such as slit fabrication tolerances, slight misalignment, or calibration inaccuracies in spatial mapping. Nonetheless, the internal consistency between all three methods strengthens the validity of the experimental results and confirms the theoretical description.

Notably, the parameter g remained remarkably stable across all methods in the Fraunhofer regime, fluctuating by less than 2,µm between the different fits. This not only indicates that the interference fringe spacing was well-resolved but also that the fitted cosine modulation component in the intensity expression accurately captured the periodicity imposed by the slit separation. Likewise, the extracted slit widths from the fits were also consistent, confirming that the sinc^2-shaped diffraction envelope appropriately modeled the diffraction effects of the finite slit width.

In the Fresnel regime, the transition from far-field to near-field behavior was clearly observed in the evolution of the recorded patterns as the slit-to-camera distance z decreased. As expected from the theoretical model based on Fresnel integrals, the fringe spacing became narrower, the oscillatory structure more complex, and the central peak more sharply defined at shorter distances. These qualitative features were well-reproduced by the numerical evaluation of the Fresnel model, where the average values of g and b from the Fraunhofer fits were held fixed. Only the overall amplitude and background offset were adjusted.

The best fits were observed for the intermediate and shortest distances, with $r^2 \approx 0.9577$ and $r^2 \approx 0.9547$, respectively, while the longest distance at $z = 42.6\,\text{cm}$ showed a slightly lower $r^2 = 0.9037$, likely due to increased compression of fringes and lower sensitivity in fringe localization. Importantly, these fits validate the theoretical Fresnel formulation as an accurate description of near-field double-slit diffraction.

Overall, the systematic progression from well-separated sinc^2-modulated $\cos^2$ patterns in the far field to more intricate near-field intensity distributions confirms the theoretical prediction of a continuous transition between Fraun-

hofer and Fresnel regimes. The data also demonstrate that both scalar diffraction models—each appropriate in their respective limits—successfully account for the observed interference and diffraction phenomena. While there is a consistent overestimation of slit parameters compared to the nominal values, the mutual agreement between independent fits, the excellent correlation coefficients, and the consistent pattern morphology across propagation distances strongly support the reliability and coherence of the experimental results.

5 Conclusion

In this experiment, the diffraction and interference of coherent light by single and double slits were investigated in both the Fraunhofer and Fresnel regimes. By systematically varying the slit geometry and the observation distance, a broad range of diffraction phenomena were explored and quantitatively analyzed using theoretical models based on scalar wave optics.

The single-slit diffraction patterns in the Fraunhofer regime were successfully fitted with the expected sinc-squared intensity distribution, allowing for accurate determination of the slit width. In the Fresnel regime, the observed deviation from the far-field pattern was well captured by the numerical evaluation of Fresnel integrals, confirming the validity of the near-field diffraction model.

For the double-slit system, three independent methods were used to extract the slit separation and width in the Fraunhofer regime: position-based analysis of minima, direct spatial fitting, and frequency-domain analysis via Fourier transformation. All methods yielded consistent values with high correlation, though the measured parameters deviated systematically from nominal experimental values. The diffraction patterns in the Fresnel regime displayed the expected evolution in fringe density and complexity with decreasing slit-to-screen distance and were modeled accurately using the Fresnel-Kirchhoff formalism.

The experiment not only confirmed the theoretical predictions of classical wave optics but also demonstrated the practical utility of diffraction patterns for precise geometric characterization of slit apertures. Furthermore, the continuous transition observed between near- and far-field regimes highlighted the conceptual coherence of optical diffraction theory and its predictive power across a wide range of physical conditions.

References

[1] M. Ziese. *O17e Diffraction.* Universität Leipzig, June 2025.

[2] M. Ziese. *Guide to Experiment O17e: Diffraction.* Universität Leipzig, June 2025.